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AIMLZC416, DSECLZC416- MFML, MFDS

Webinar: #1

AGENDA



Problems on

- System of linear equations
- Vector spaces
- Inner Product Space
- Cholesky Factorization

PROBLEM 1



You are given a system of three linear equations as shown below:

$$\alpha x + 2y + 6z = b_1$$

$$\alpha x + \alpha y + 5z = b_2$$

$$\alpha x + \alpha y + \alpha z = b_3$$

In each of the equations, you can observe the presence of an unknown constant α . Derive the three possible values of α for which the given linear system will fail to have three pivots. Provide a valid reason in each case.

Consider the coefficient matrix

$$\mathbf{A} = \begin{bmatrix} \alpha & 2 & 6 \\ \alpha & \alpha & 5 \\ \alpha & \alpha & \alpha \end{bmatrix}.$$

To convert it into Row Echelon Form (REF), perform row operations:

$$\mathbf{A} \xrightarrow{R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1} \begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & \alpha - 2 & \alpha - 6 \end{bmatrix}.$$

Continue row reduction:

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & 0 & \alpha - 5 \end{bmatrix}.$$

Hence, the REF of **A** is

$$\begin{bmatrix} \alpha & 2 & 6 \\ 0 & \alpha - 2 & -1 \\ 0 & 0 & \alpha - 5 \end{bmatrix}.$$

Clearly, **A** will not have 3 pivots if

$$\alpha = 0, \quad \text{or} \quad \alpha - 2 = 0, \quad \text{or} \quad \alpha - 5 = 0.$$

$$\boxed{\alpha = 0, 2, 5}.$$

PROBLEM 2



$$\text{Let } \mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

- (a) Using elementary row operations, write the matrix in its REF.
- (b) Let $\mathbf{V}$ be a vector subspace spanned by the columns of matrix $\mathbf{A}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis?
- (c) Let $\mathbf{V}$ be a vector subspace spanned by vectors $\mathbf{x}$, such that $\mathbf{Ax} = \mathbf{0}$. Give a set of vectors that form the basis for the vector space $\mathbf{V}$. What is the number of vectors in this basis?
- (d) Give the set of linearly independent rows of $\mathbf{A}$. What is the number of vectors in this set?

SOLUTION



(a)

$$\mathbf{A} = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

$$\xrightarrow{R_2 \leftarrow R_2 + 3 * R_1, R_3 \leftarrow R_3 - 2 * R_1, R_2 \leftarrow R_2 / 5, R_3 \leftarrow R_3 - R_2, R_1 \leftarrow R_1 - 2 * R_2}$$

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

NOTE: REF is not unique.

(b) column number 1 and column number 3 have pivot.
Basis of column space is:

$$\begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$$

Rank of col space = 2

(c) From REF, the null space is:

$$\begin{bmatrix} x \\ r \\ y \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} r + \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix} s + \begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} t$$

Rank of null space = 3

(d) From REF, row 1 and row 2 form basis of row space.

Rank of row space = 2

PROBLEM 3



Suppose $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$, where $\mathbf{C}_m \in \mathbb{R}^4$, $m = 1, 2, 3, 4$ are columns of $\mathbf{A}$. It is known that $\text{rank}(\mathbf{A}) = 2$ and $\mathbf{C}_2 = 3\mathbf{C}_1$ and $\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3$. If a particular solution of $\mathbf{Ax} = \mathbf{b}$ is $[1, 0, 1, 0]^T$, then

- (a) Find the general solution of $\mathbf{Ax} = \mathbf{b}$.
- (b) Find $\mathbf{b}$ if $\text{RREF}(\mathbf{A}) = \mathbf{A}$.

SOLUTION



(a) Now $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4]$ where $\mathbf{C}_m \in \mathbb{R}^4$, $m = 1, 2, 3, 4$ are columns of $\mathbf{A}$. It is known that $\text{rank}(\mathbf{A}) = 2$

$$\mathbf{C}_2 = 3\mathbf{C}_1 \Rightarrow 3\mathbf{C}_1 - \mathbf{C}_2 + 0\mathbf{C}_3 + 0\mathbf{C}_4 = \mathbf{0}$$

$$\Rightarrow [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} = \mathbf{0}$$

$$\mathbf{C}_4 = 2\mathbf{C}_1 + 3\mathbf{C}_3 \Rightarrow 2\mathbf{C}_1 + 0\mathbf{C}_2 + 3\mathbf{C}_3 - 1\mathbf{C}_4 = \mathbf{0}$$

$$\Rightarrow [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} = \mathbf{0}$$

Thus, the general solution of $\mathbf{Ax} = \mathbf{b}$ is given by

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} 3 \\ -1 \\ 0 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 2 \\ 0 \\ 3 \\ -1 \end{bmatrix} \mid \forall \lambda_1, \lambda_2 \in \mathbb{R} \right\}$$

(b) $\mathbf{A} = [\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \mathbf{C}_4] = [\mathbf{C}_1, 3\mathbf{C}_1, \mathbf{C}_3, 2\mathbf{C}_1 + 3\mathbf{C}_3]$ and $\text{rank}(\mathbf{A}) = 2$. Therefore $\mathbf{C}_1, \mathbf{C}_3$ are pivot columns & $\text{RREF}(\mathbf{A}) = \mathbf{A}$.

$$\Rightarrow \mathbf{C}_1 = [1, 0, 0, 0]^T \text{ and } \mathbf{C}_3 = [0, 1, 0, 0]^T$$

$$\Rightarrow \mathbf{C}_2 = [3, 0, 0, 0]^T \text{ and } \mathbf{C}_4 = [2, 3, 0, 0]^T$$

$$\Rightarrow \mathbf{A} = \begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \text{ Therefore } \mathbf{b} = \mathbf{A}[1, 0, 1, 0]^T = [1, 1, 0, 0]^T$$

PROBLEM 4



Consider $M = \{\mathbf{A} \in \mathbb{R}^{2 \times 2} \mid \mathbf{A} = -\mathbf{A}^T\}$.

- (a) Prove that M is a subspace of vector space V , where V is set of all 2×2 real matrices.
- (b) Prove or disprove $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis for M defined in above.
- (c) Prove or disprove that $\left\{ \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} \right\}$ is a linearly independent set.

SOLUTION



(a) Clearly the zero matrix of order 2, $\mathbf{0} \in M \Rightarrow M \neq \phi$.

Let $\mathbf{A}, \mathbf{B} \in M, k \in \mathbb{R}$

$$\Rightarrow \mathbf{A} = -\mathbf{A}^T, \mathbf{B} = -\mathbf{B}^T$$

$$\Rightarrow \mathbf{A} + \mathbf{B} = -\mathbf{A}^T - \mathbf{B}^T = -(\mathbf{A} + \mathbf{B})^T$$

$$\Rightarrow \mathbf{A} + \mathbf{B} \in M$$

$$k\mathbf{A} = k(-\mathbf{A}^T) = (-k)\mathbf{A}^T = -(k\mathbf{A})^T$$

So, M is a subspace.

(b) Clearly $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ does not belong to M and hence cannot be a basis of M .

(c) Consider $a \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix} + b \begin{bmatrix} 5 & -5 \\ 5 & 5 \end{bmatrix} + c \begin{bmatrix} 0 & -3 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

$$\Rightarrow a + 5b = 0$$

$$-a - 5b - 3c = 0$$

$$a + 5b + 3c = 0$$

$$5b + c = 0$$

$\Rightarrow a = b = c = 0$. Hence it a linearly independent set.

PROBLEM 5

Let $\mathbf{A} = \begin{bmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$.

Then define $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T \mathbf{A} \mathbf{y}$ where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^4$.

- (a) Prove that $\langle \cdot, \cdot \rangle_{\mathbf{A}}$ is an inner product on $\mathbb{R}^4$.
- (b) Find all possible $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ such that $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$ where $\mathbf{x} = [1, 0, 0, 0]^T$

(a) Clearly $\mathbf{A} = \mathbf{A}^T$ and hence $\mathbf{A}$ is symmetric. Now

$$\begin{aligned}\mathbf{x}^T \mathbf{A} \mathbf{x} &= 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2 - 2x_3x_4 + 2x_4^2 \\ &= x_1^2 + (x_1 - x_2)^2 + (x_2 - x_3)^2 + (x_3 - x_4)^2 + x_4^2 > 0\end{aligned}$$

if $\mathbf{x} \neq 0$.

Thus $\mathbf{A}$ is symmetric positive definite. Hence by theorem $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = \mathbf{x}^T \mathbf{A} \mathbf{y}$ is an inner product.

(b) If $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ and $\mathbf{x} = [1, 0, 0, 0]^T$

we want $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$

$$\Rightarrow 2y_1 - y_2 = 0 \Rightarrow 2y_1 = y_2$$

Thus $\mathbf{y} = [y_1, y_2, y_3, y_4]^T$ with $2y_1 = y_2$, $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathbf{A}} = 0$.

PROBLEM 6



Consider the positive definite matrix $A \in \mathbb{R}^{3 \times 3}$ defined as:

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

- (i) Derive a lower triangular matrix L so that A can be factorized as $A = LL^T$. Derive all the three eigenvalues of L .
- (ii) Let $x \in \mathbb{R}^3$ be a nonzero vector. Define $z = L^T x$ where $L \in \mathbb{R}^{3 \times 3}$ satisfies the equality $A = LL^T$. Prove that z can never be the zero vector.

SOLUTION



Given

$$A = \begin{bmatrix} 9 & 6 & 3 \\ 6 & 8 & 12 \\ 3 & 12 & 75 \end{bmatrix}$$

Since A is positive definite, it admits a Cholesky factorization $A = LL^T$ where L is lower triangular.

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

Then

$$LL^T = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{21}l_{11} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{31}l_{11} & l_{31}l_{21} + l_{32}l_{22} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

Equating with A:

$$l_{11}^2 = 9 \Rightarrow l_{11} = 3$$

$$3l_{21} = 6 \Rightarrow l_{21} = 2$$

$$3l_{31} = 3 \Rightarrow l_{31} = 1$$

$$l_{21}^2 + l_{22}^2 = 8 \Rightarrow 4 + l_{22}^2 = 8 \Rightarrow l_{22} = 2$$

$$l_{21}l_{31} + l_{22}l_{32} = 12 \Rightarrow 2 + 2l_{32} = 12 \Rightarrow l_{32} = 5$$

$$l_{31}^2 + l_{32}^2 + l_{33}^2 = 75 \Rightarrow 1 + 25 + l_{33}^2 = 75 \Rightarrow l_{33} = 7$$

$$L = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 1 & 5 & 7 \end{bmatrix}$$

Since L is triangular, its eigenvalues are its diagonal entries:

$$\lambda_1 = 3, \quad \lambda_2 = 2, \quad \lambda_3 = 7$$

(ii) Let $x \in \mathbb{R}^3$ be nonzero and define $z = L^T x$. Then

$$x^T A x = x^T L L^T x = (L^T x)^T (L^T x) = z^T z = \|z\|^2$$

Since A is positive definite,

$$x^T A x > 0 \quad \forall x \neq 0$$

Hence

$$\|z\|^2 > 0 \Rightarrow z \neq 0$$

$z = L^T x$ can never be the zero vector if $x \neq 0$.

PROBLEM 7



- (a) Assume that you are given a matrix $A \in \mathbb{R}^{n \times n}$ where $A = [a_{ij}]$ ($1 \leq i, j \leq n$). The matrix A has the following interesting property:

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for each of } i = 1, 2, \dots, n.$$

- (i) Can zero be one of the eigenvalues of this matrix? Give reasoning for your answer.
- (ii) Derive the determinant of this matrix.
- (iii) Derive at least one eigenvector–eigenvalue pair of this matrix.

SOLUTION



(a) Given $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ such that

$$a_{i1} + a_{i2} + \cdots + a_{in} = 0 \quad \text{for each } i = 1, 2, \dots, n.$$

(i) Let

$$\mathbf{e} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

Then

$$A\mathbf{e} = \begin{bmatrix} a_{11} + a_{12} + \cdots + a_{1n} \\ a_{21} + a_{22} + \cdots + a_{2n} \\ \vdots \\ a_{n1} + a_{n2} + \cdots + a_{nn} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

SOLUTION



Hence

$$A\mathbf{e} = 0\mathbf{e}.$$

0 is an eigenvalue of A .

(ii) Since 0 is an eigenvalue, A is singular. Therefore,

$$\det(A) = 0.$$

(iii) An eigenvalue–eigenvector pair is

$$\lambda = 0, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

PROBLEM 8



Consider the data matrix given below

$$C = \begin{bmatrix} -5 & 10 & -5 & 5 \\ 8 & -16 & -4 & -20 \\ 4 & -8 & 5 & -3 \\ -3 & 6 & 5 & 11 \end{bmatrix}.$$

- (i) Reduce the matrix C to its reduced row echelon form (RREF).
- (ii) Find a basis for the subspace spanned by the columns of the matrix C .
- (iii) Suppose that C is the augmented matrix of a system of linear equations $A\mathbf{x} = \mathbf{b}$. Determine whether the system is consistent. If it is consistent, find its general solution.

b) If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^4$ and $\mathbf{v}_4$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, then prove or disprove that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

forms a linearly independent set.

c) If $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4 \in \mathbb{R}^3$ and $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set, then prove or disprove that

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$$

forms a linearly dependent set.

SOLUTION



$$\text{i) } R_1 \leftarrow \frac{R_1}{-5} : \begin{bmatrix} 1 & -2 & 1 & -1 \\ 8 & -16 & -4 & -20 \\ 4 & -8 & 5 & -3 \\ -3 & 6 & 5 & 11 \end{bmatrix}$$

$$R_2 \leftarrow R_2 - 8R_1, R_3 \leftarrow R_3 - 4R_1, R_4 \leftarrow R_4 + 3R_1, R_2 \leftarrow \frac{R_2}{-12}$$

$$\begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 8 & 8 \end{bmatrix}$$

$$R_1 \leftarrow R_1 - R_2, R_3 \leftarrow R_3 - R_2, R_4 \leftarrow R_4 - 8R_2 : \begin{bmatrix} 1 & -2 & 0 & -2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ii) Since, column 1, 3 are pivot columns,

$$\text{Basis} = \{[-5, 8, 4, -3]^T, [-5, -4, 5, 5]^T\}$$

iii) Given $C = [A \mid b]$, $\Rightarrow \text{rank}(C) = \text{rank}[A \mid b]$.

Clearly, from (i),

$$\text{rank}(A) = \text{rank}[A \mid b] = 2.$$

i.e., System is consistent.

Since column 2 is a non-pivot column, x_2 is a free variable. Let $x_2 = t$.

From row 2 of $\text{rref}[A \mid b]$, we get $x_3 = 1$. From row 1, $x_1 - 2x_2 = -2 \Rightarrow x_1 = -2 + 2t$.

The general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + t \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \quad \forall t \in \mathbb{R}.$$

b) False, as $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ can be a linearly dependent set, making $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ a linearly dependent set.

c) Yes, as $\mathbb{R}^3$ has dimension 3, so any set containing more than 3 vectors is linearly dependent.

PROBLEM 9



a) Define

$$\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^T A \mathbf{y} \quad \text{where } \mathbf{x}, \mathbf{y} \in \mathbb{R}^2 \quad \text{and} \quad A = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}.$$

- (i) Prove $\langle \cdot, \cdot \rangle_A$ is an inner product on $\mathbb{R}^2$.
- (ii) Find $\|\mathbf{x}\|_A$ where $\|\cdot\|_A$ is the norm induced by the inner product defined above for $\mathbf{x} = [1, 2]^T$.

b) Let A be a 2×2 symmetric real matrix with trace of A equal to 0. Can A be a symmetric positive definite matrix? Justify your answer.

c) A professor gave his student the following matrix

$$A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}.$$

Help the student to find a lower triangular matrix L such that

$$A = LL^T.$$

a) i) Clearly $A = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} = A^T$

And $\det A_{11} = \det[4] = 4 > 0$, $\det A = 24 - 4 = 20 > 0$

$\therefore A$ is symmetric and positive definite

$\Rightarrow \langle x, y \rangle_A = x^T A y$ is an inner product on $\mathbb{R}^2$.

ii) Note that $\langle x, x \rangle_A = [1, 2] \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$= [1, 2] \begin{bmatrix} 8 \\ 14 \end{bmatrix} = 8 + 28 = 36$$

$$\therefore \|x\|_A = \sqrt{\langle x, x \rangle_A} = \sqrt{36} = 6$$

b) Trace of A = sum of eigenvalues = 0.

Clearly, both eigenvalues are real due to the spectral theorem. If one is positive, the other eigenvalue has to be negative, as their sum is zero.

$\therefore$ Eigenvalues are not strictly positive.

$\Rightarrow$ It is not positive definite.

c) Let $L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$

$$\Rightarrow LL^T = \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{21}l_{31} + l_{22}l_{32} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}$$

$$4 = \ell_{11}^2 \Rightarrow \ell_{11} = 2, \quad 1 = \ell_{11}\ell_{21} \Rightarrow \ell_{21} = 0.5.$$

$$1 = \ell_{11}\ell_{31} \Rightarrow \ell_{31} = 0.5$$

$$3 = \ell_{21}^2 + \ell_{22}^2 \Rightarrow \ell_{22} = \sqrt{3 - 0.25} = \sqrt{2.75} = 1.6583$$

$$1 = 0.5(0.5) + \sqrt{2.75}\ell_{32} \Rightarrow \ell_{32} = \frac{0.75}{\sqrt{2.75}} = 0.4523$$

$$2 = 0.25 + \frac{(0.75)^2}{2.75} + \ell_{33}^2 \Rightarrow \ell_{33} = \sqrt{1.75 - \frac{(0.75)^2}{2.75}} = 1.2432$$

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THANK YOU